

Curate Software Design Document (SDD)

1. Document Information

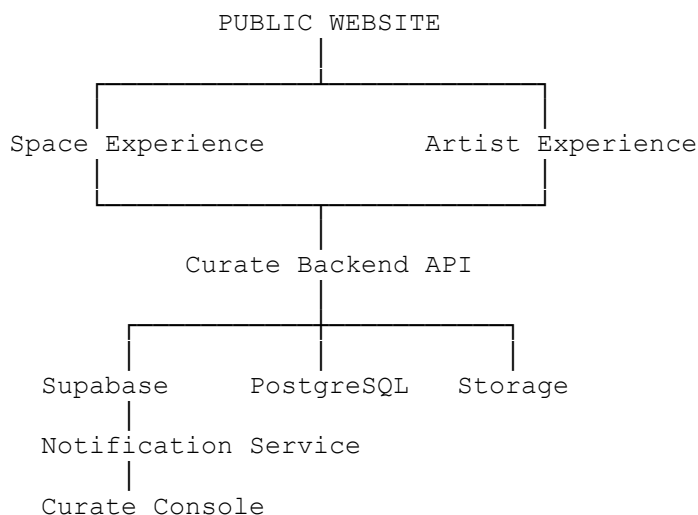
- **Project:** Curate
 - **Version:** 1.0 (MVP)
 - **Document Type:** Software Design Document
 - **Architecture:** React + Express + Supabase
 - **Status:** Design Phase
-

2. Purpose

This document defines the complete software architecture, user workflows, system interactions, data flow, authentication model, backend services, and module communication for Curate.

It serves as the implementation blueprint for developers and should remain consistent with the approved UI/UX.

3. System Overview



4. System Architecture

Client (React)

↓

REST API (Express)

↓

Business Logic Layer

↓

Supabase Services

↓

PostgreSQL

↓

Supabase Storage

5. Modules

Module 1

Public Website

Purpose

Marketing

Contains

- Home
- Spaces
- Gallery
- Artists
- About
- Let's Talk

Users

Guest

Authenticated Users

Module 2

Space Experience

Purpose

Manage subscribed spaces.

Contains

Authentication



Register Space



Dashboard



Browse Collections



Artwork Details



Frame Selection



Payment



Tracking



Rotation



Invoices



Support

Module 3

Artist Experience

Purpose

Manage artist submissions.

Contains

Authentication

↓

Profile

↓

Upload

↓

Review

↓

Workspace

↓

Portfolio

↓

Notifications

↓

Installations

↓

Payouts

Module 4

Curate Console

Purpose

Internal operations.

Contains

CEO

Operations

Accounts

Moderation

Printing

Payments

Reports

6. User Roles

Guest

↓

Space Owner

↓

Artist

↓

Internal Staff

├─ CEO

├─ Manager

├─ Accounts

└─ Operations

7. Authentication Flow

User

↓

Supabase Auth

↓

Email Verification

↓

Role Assignment

↓

JWT Session

↓

Protected Routes

8. Space Owner Flow

Visitor

↓

Spaces

↓

Sign In

↓

Register Space

↓

Verification

↓

Dashboard

↓

Browse Collections

↓

Artwork Details

↓

Frame Selection

↓

Checkout

↓

QR Payment

↓

Verification

↓

Payment Success

↓

Track Installation

↓

Rotation

↓

Invoice

9. Artist Flow

Visitor

↓

Artists

↓

Sign In

↓

Create Profile

↓

Upload Photograph

↓

Validation

↓

Pending Review

↓

Approved

↓

Workspace

↓

Notification

↓

Artwork Selected

↓

Installation

↓

Payout

10. Payment Flow

Current MVP

Checkout

↓

Generate Dynamic QR

↓

User Pays

↓

Payment Verification

↓

Success

↓

Order Created

Failure

Checkout

↓

QR

↓

Payment Failed

↓

Retry

↓

Cancel

Future

Payment Service

↓

Stripe

or

Razorpay

Only the service changes.

The workflow stays identical.

11. Upload Flow

Upload

↓

Client Validation

↓

Base64 Encoding

↓

Async Upload

↓

Supabase Storage

↓

Public URL

↓

Database Entry

↓

Review Queue

↓

Published

12. Notification Flow

Action

↓

Notification Service

↓

Database

↓

In-App Notification

↓

Read

↓

Archived

Events

- Artwork Selected
 - Upload Approved
 - Payment Confirmed
 - Installation Scheduled
 - Rotation Reminder
-

13. Order Flow

Payment

↓

Order

↓
Printing
↓
Framing
↓
Installation
↓
Completed
↓
Rotation Queue

14. Rotation Flow

Current Collection
↓
Rotation Due
↓
Curate Review
↓
New Collection
↓
Owner Approval
↓
Installation
↓
History Updated

15. Storage Structure

Supabase Storage

/artworks

/profiles

/spaces

/invoices

/documents

/thumbnails

16. Database Modules

Authentication

Users

↓

Profiles

↓

Spaces

↓

Artworks

↓

Orders

↓

Frames

↓

Payments

↓

Notifications

↓

Rotation

↓

Invoices

17. API Modules

/auth

/users

/spaces

/artworks

/orders

/payments

/uploads

/notifications

/rotation

/invoices

/admin

18. Security

Supabase Authentication

↓

Role Validation

↓

API Authorization

↓

Database Policies

↓

Storage Policies

19. Logging

Every important action

↓

Activity Log

↓

Audit Log

↓

Error Log

20. System States

Guest

Authenticated

Pending Verification

Verified

Order Pending

Printing

Framing

Installation

Completed

Cancelled

21. Future Services

AI Recommendation

Search

Chatbot

Stripe

SMS

WhatsApp

Analytics

22. Folder Structure

client/

server/

shared/

components/

features/

hooks/

services/

contexts/

routes/

middleware/

database/

storage/

utils/

types/

23. Sequence Diagram

Space Owner

↓

Login

↓

Browse Collection

↓

Select Artwork

↓

Payment

↓

Order

↓

Printing

↓

Installation

↓

Notification

↓

Rotation

Artist

Login

↓

Upload

↓

Review

↓

Selected

↓

Installation

↓

Payment

↓

Notification

24. MVP Milestones

Phase 1

- Public Website

Phase 2

- Space Experience

Phase 3

- Artist Experience

Phase 4

- Curate Console

Phase 5

- AI & Automation